Tomáš Plšek

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Scientific Education

2021 – 2026	**PhD:** *Investigating the Physics of Hot Galactic Atmospheres*, Department of Theoretical Physics and Astrophysics, Masaryk University, Brno, CZ
2019 – 2021	**MSc:** *The relation between Bondi accretion rate and jet power in giant elliptical galaxies*, Department of Theoretical Physics and Astrophysics, Masaryk University, Brno, CZ
2016 – 2019	**BSc:** *Dark matter modelling of galaxy clusters*, Department of Theoretical Physics and Astrophysics, Masaryk University, Brno, CZ

Work Experience

since 2021	**Researcher** at Department of Theoretical Physics and Astrophysics, Masaryk University, Brno, CZ
2019 – 2020	**Researcher** at Department of Physical Electronics, Masaryk University, Brno, CZ
2017 – 2018	**Astronomical Demonstrator** at Vyškov Observatory, CZ

Honors & Awards

2022	**Dean's Award 2022**, Faculty of Science, Masaryk University, Brno

Supervision & Co-supervision

since 2025	**Consultant:** *ML-powered detection of accretion shocks in IllustrisTNG simulation*, Master's thesis by Barbora Miklošová, Masaryk University, Brno
since 2024	**Consultant:** *Search for galaxy clusters in XMM data of the Euclid Fornax Deep Field*, Master's thesis by Martin Kolář, Masaryk University, Brno
2025	**Supervisor:** *Probing the radio-mechanical AGN feedback using multi-cavity systems*, Bachelor's thesis by Barbora Miklošová, Masaryk University, Brno
2024	**Consultant:** *X-ray spectroscopy of the cosmic web*, Bachelor's thesis by Martin Kolář, Masaryk University, Brno
2023	**Consultant:** *Machine learning enhanced search for transient events in light curves*, Master's thesis by Bc. Martin Chobola, Masaryk University, Brno
2023	**Consultant:** *Studying the brightest galaxies in the epoch of reionisation*, Master's thesis by Klaudia Protušová, Masaryk University, Brno
2023	**Consultant:** *Studying the heating/cooling balance in galaxy clusters with the Athena X-ray observatory*, Master's thesis by Eva Baťková, Masaryk University, Brno

Teaching & Scientific Outreach

2024 – 2025	**Astronomical camp group leader** (Astronomická expedice)
Jan 26 2022	**Guest speaker:** 'Talks about Cosmos' (Rozhovory o Vesmíru)
2023 – 2026	**Teaching assistance:** F4500 Python for Physicists @MUNI
2024 – 2026	**Teaching assistance:** F6550 Structure and Evolution of the Universe @MUNI
2022 – 2024	**Teaching assistance:** F9888 High Energy Astrophysics @MUNI
2021 – 2024	**Teaching assistance:** F5611 Introduction to Machine Learning for Astronomers @MUNI

Selected Conference Talks

Jun 27 2025	**Conference:** *Cavity Detection Tool (CADET)*, EAS 2025, Cork, Ireland
Jun 25 2025	**Conference:** *The Abundance Drop in the Core of the Centaurus cluster*, EAS 2025, Cork, Ireland
Oct 16 2024	**Seminar:** *Cavity Detection Tool*, Virtual Astronomy Software Talks (VAST)
Sep 19 2024	**Conference:** *Studying radio-mechanical AGN feedback with X-ray cavities*, GALCROSS, Brno, Czech Republic
Jun 4 2024	**Conference:** *Studying radio-mechanical AGN feedback with X-ray cavities*, Galactic Nuclei in the Cosmological Context, University of Szczecin, Poland
Jul 11 2023	**Seminar:** *Cavity Detection Tool (CADET)*, seminar of XOC group, Stanford University, USA
Jul 11 2023	**Seminar:** *The relation between accretion rate, black hole mass, and jet power in massive early-type galaxies*, KIPAC Tea Talk, Stanford University, USA
Jul 2 2023	**Workshop:** *Cavity Detection Tool (CADET)*, High Resolution X-ray Spectroscopy Workshop, MIT, Cambridge, USA
Mar 25 2023	**Workshop:** *Cavity Detection Tool (CADET)*, AHEAD2020 High Resolution X-ray Spectroscopy School, Alicante, Spain
Nov 22 2022	**Seminar:** *The relation between accretion rate, black hole mass, and jet power in massive early-type galaxies*, Institute of Physics, Silesian University, Opava, Czech Republic
Aug 15 2022	**Workshop:** *Cavity Detection Tool*, 6th ICM Theory and Computation Workshop, Niels Bohr Institute, Copenhagen, Denmark
Jun 3 2022	**Conference:** *Relation between accretion rate and jet power in early-type galaxies*, Cologne-Prague-Brno meeting, Brno, Czech Republic

Conference organization

Aug 20-24 2026	**LOC:** 8th ICM Theory and Computation Workshop, Brno, Czech Republic
May 18-22 2026	**LOC:** IAU Symposium: Traversing the Galactic Center in Space and Time, Brno, Czech Republic
Sep 16-20 2024	**LOC:** Galaxies at Crossroads: Outflows and IMF in the VLT/ELT/ALMA/JWST Era, Brno, Czech Republic
Sep 6-8 2022	**LOC:** Monitoring the High-Energy Sky with Small Satellites, Brno, Czech Republic
Jun 1-3 2022	**LOC:** Cologne Prague Brno meeting, Brno, Czech Republic

Involvement in the scientific community

- **Member of the XRISM Collaboration**

  I am an external member of the Centaurus target team. I am involved in several XRISM Collaboration published or submitted publications, which utilise PV+GO Resolve observations of the core of the Centaurus cluster, where I was responsible for the complementary analysis of *Chandra* data.

  I have actively participated in XRISM Science Team meetings (March 2024 Tokyo, Sep 2024 Tokyo, Feb 2025 Phoenix, Jul 2025 Leiden).

- **Member of the Athena collaboration**

  I am involved in testing the capabilities of the *NewAthena/XIFU* detector for spatially mapping the cold gas and the heating / cooling balance of hot intracluster medium (ICM).

Programming & Software Skills

- **Python:** Anaconda, Matplotlib, Numpy, Scipy, Astropy, Pandas, Jupyter
- **Machine learning:** Scikit-learn, Keras, Tensorflow, PyTorch, JAX
- **Bayesian analysis:** PyMC3, UltraNest
- **X-ray data analysis:** CIAO, HEASoft, SAOImageDS9
- **X-ray spectral fitting:** Xspec, SPEX, Sherpa, BXA
- **X-ray simulators:** PyXsim, SOXS, SIXTE, MARX
- **Cosmological simulations:** Gadget-2, IllustrisTNG data
- **Linux:** Bash scripting, SSH, git
- **Text & Web:** LaTeX, Markdown, HTML
- **AI & LLM tools:** Ollama (open-weight models), Warp, Claude Code